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IMDb Top 1000 Movies Dataset Analysis

Introduction

The IMDb Top 1000 movies dataset offers a comprehensive view into the trends and characteristics of highly rated films. Comprising variables such as IMDb ratings, number of votes, gross earnings, runtime, and meta scores, this dataset serves as a valuable tool for analyzing what factors contribute to a film's success. The motivation behind this analysis lies in identifying patterns between audience engagement, financial performance, and film characteristics. Such insights are beneficial for producers, marketers, and investors seeking to optimize content performance and appeal.

This report investigates three primary research questions: (1) How does IMDb rating correlate with the number of votes and gross earnings? (2) Can we classify movies using k-Nearest Neighbors based on IMDb ratings? (3) How does a movie's runtime impact its IMDb rating and gross earnings?

Question 1: How does IMDb Rating Correlate with the Number of Votes and Gross Earnings?

To examine the relationships between IMDb rating, number of votes, and gross earnings, a correlation matrix was constructed. A moderate positive correlation between IMDb rating and the number of votes suggests that more popular films tend to have slightly higher ratings. However, the correlation between ratings and gross earnings is weak, indicating that a film's box office performance does not necessarily align with audience or critical reception. Conversely, the number of votes and gross earnings show a moderately positive correlation, supporting the idea that widely viewed films tend to generate more revenue.

Figure 1: Correlation Heatmap of IMDb Rating, Votes, and Gross

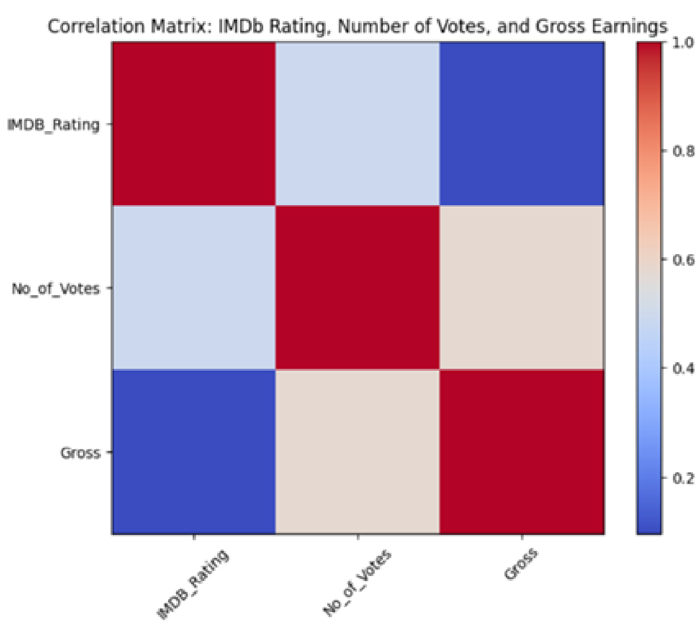

Figure 1-This heatmap visualizes the correlation coefficients between three key variables in the film industry. Each square in the matrix indicates the strength and direction of a linear relationship between two variables, with values ranging from ***-1 (perfect negative correlation)*** *to* ***+1 (perfect positive correlation)****. The color scale on the right helps interpret the values, where:*

***Red** indicates strong positive correlation, **Blue** indicates strong negative correlation, **Neutral tones** (light colors) suggest weak or no correlation.:*

Figure 2: IMDb Rating vs. Number of Votes (with Regression Line)

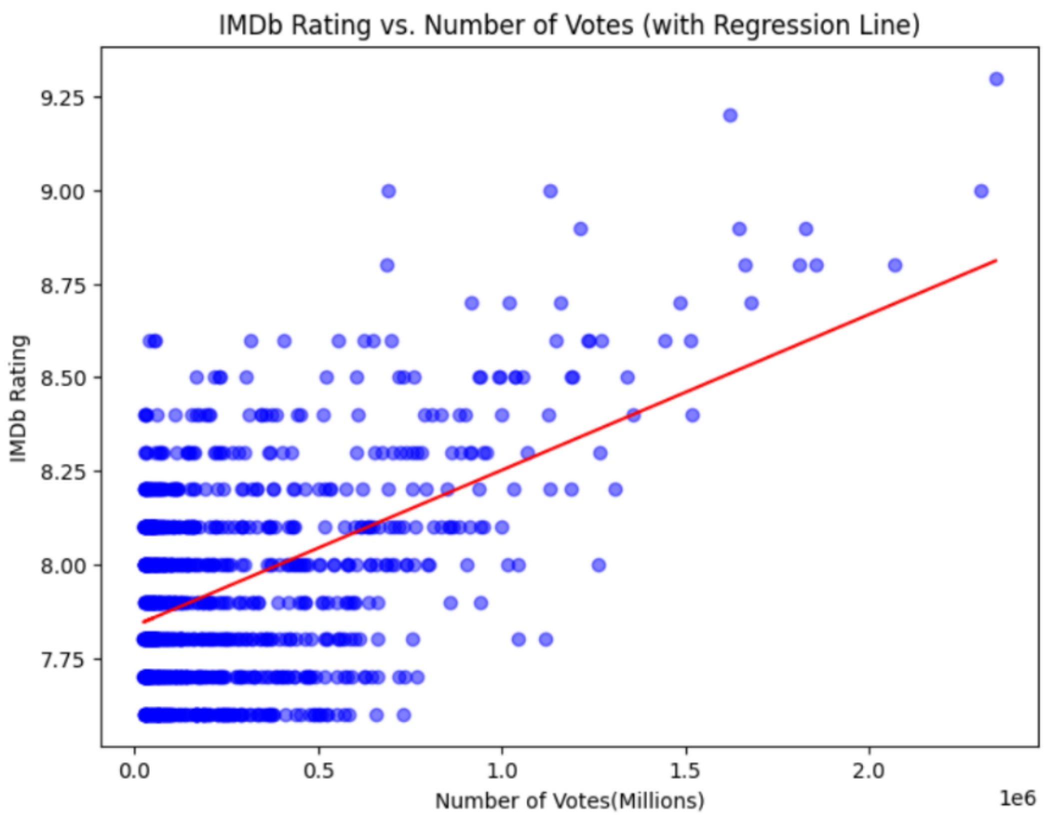

Figure 2-This scatter plot shows the relationship between a movie's IMDb rating and its number of votes. The red regression line indicates a positive correlation, as the number of votes increases, the IMDb rating tends to increase slightly

The scatterplot and regression line show a slight upward trend, indicating that IMDb ratings generally increase with the number of votes. This suggests that higher audience engagement may correlate with better ratings, possibly due to cult followings or broader appeal. While ratings vary more at lower vote counts, films with significant vote totals often exceed an 8.0 rating.

Figure 3: IMDb Rating vs. Gross Earnings (with Regression Line)

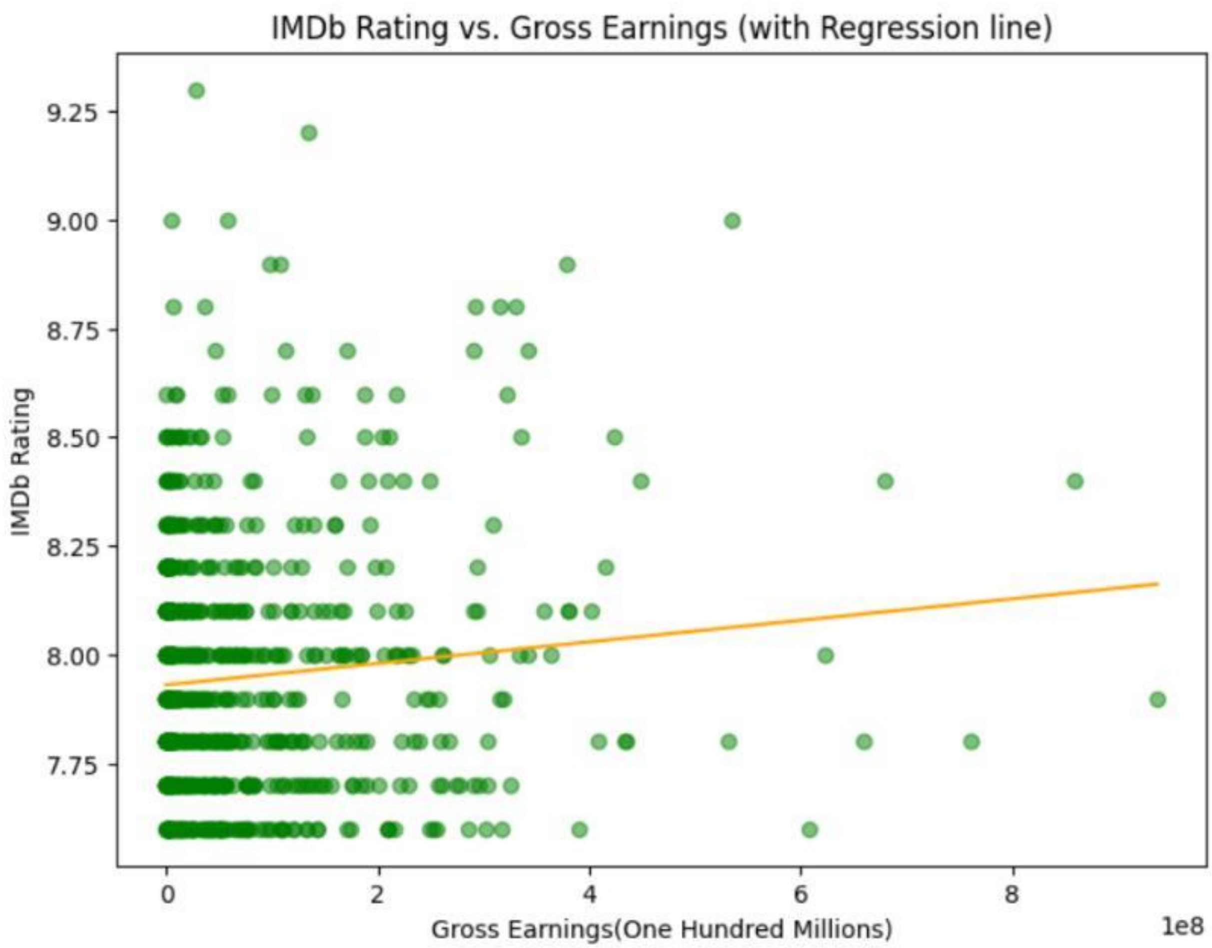

Figure 3-This scatter plot compares IMDb rating to gross earnings. The orange regression line is relatively flat, suggesting a weak positive correlation between a film's rating and its earnings.

In this scatterplot, the regression line shows only a gentle upward slope, indicating a weak relationship between IMDb ratings and gross earnings. This suggests that high-grossing films do not always receive high ratings, highlighting the influence of non-quality-related factors such as marketing, star power, or franchise appeal on earnings.

Question 2: Can We Classify Movies Based on IMDb Ratings Using k-Nearest Neighbors?

To explore the potential of classifying movies based on their IMDb ratings, a machine learning model using the k-Nearest Neighbors (k-NN) algorithm was applied. IMDb ratings were binned into three categories: High (≥ 8.4), Medium (7.9–8.4), and Low (< 7.9). The model used the following features for classification: runtime, Metascore, number of votes, and gross earnings.

The k-NN algorithm determines the class of a new movie by examining the categories of the five closest movies in the dataset (n_neighbors = 5) based on those features. If most of these neighbors are rated 'Medium,' for example, the new movie will likely be classified as 'Medium' as well.

The classifier successfully identified 46 movies as 'Low,' 38 as 'Medium,' and 9 as 'High.' Misclassifications occurred mainly in the Medium group, with 25 movies misclassified as 'Low' and 6 as 'High.' Six High-rated movies were classified as 'Medium,' but none were misclassified as 'Low.' This pattern reveals that the model handles extremes better than mid-range cases, likely due to overlapping feature distributions and possible class imbalance.

Figure 4: k-NN Classification Results

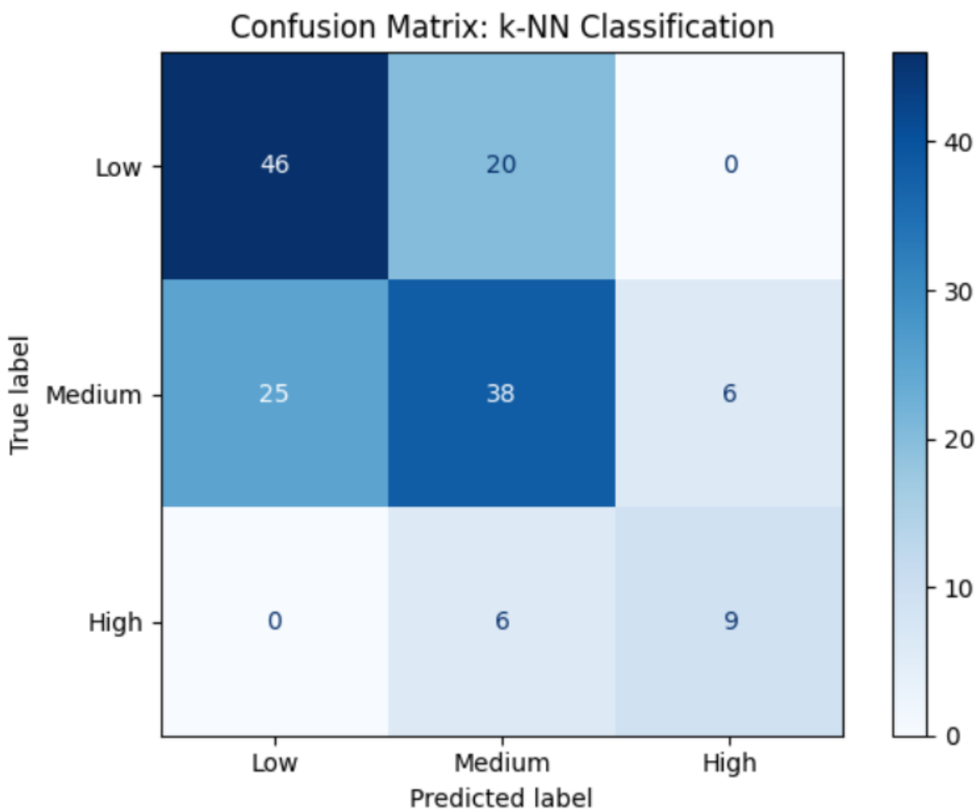

Figure 4-This matrix shows the performance of a k-Nearest Neighbors (k-NN) classifier used to predict movie categories (Low, Medium, High) based on some features (likely including rating, votes, or gross).

Figure 5: IMDb Rating Distribution

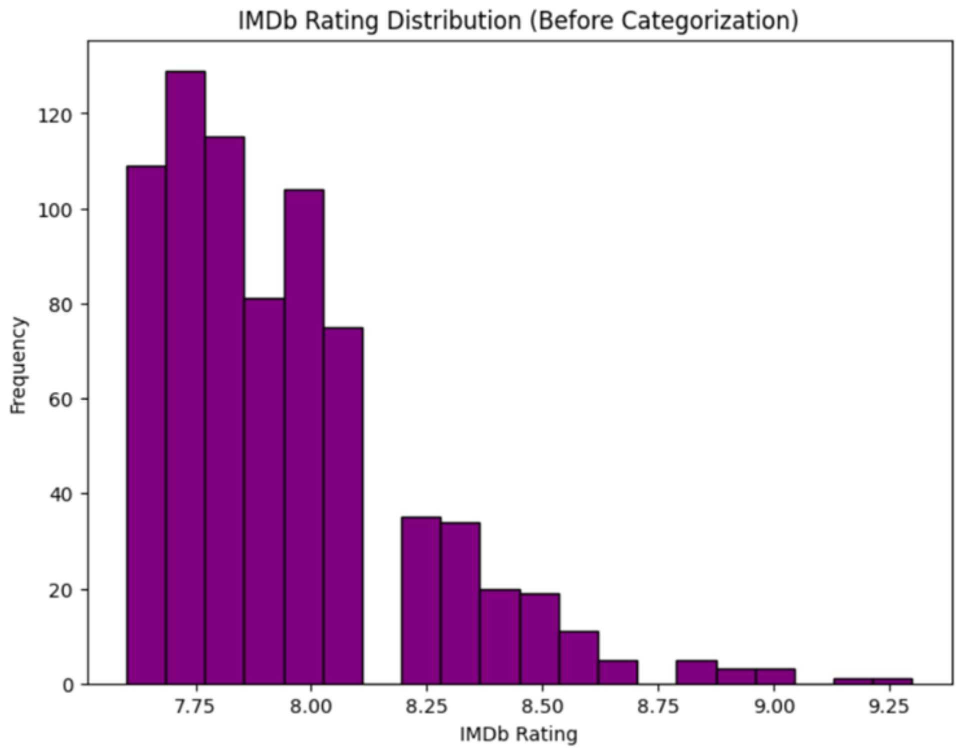

Figure 5-This is a histogram showing the distribution of IMDb ratings for a dataset of movies or shows.

The distribution of ratings reveals a strong concentration between 7.6 and 8.2, producing a left-skewed histogram. The lack of truly 'Low' ratings creates challenges for the classifier, as the classes are not evenly distributed. This class imbalance may explain why the model struggles to accurately classify Medium-rated films.

Question 3: How Does a Movie's Runtime Impact Gross Earnings?

To determine whether runtime impacts financial success, a scatterplot of runtime versus gross earnings was analyzed. While a weak positive trend is present, the wide variance and presence of outliers suggest runtime alone is not a reliable predictor of gross earnings. Most runtimes fall between 80 and 150 minutes, but earnings fluctuate significantly, indicating other variable, such as genre, cast, budget, or marketing, may play larger roles.

Figure 6: Runtime vs. Gross Earnings (with Regression Line)

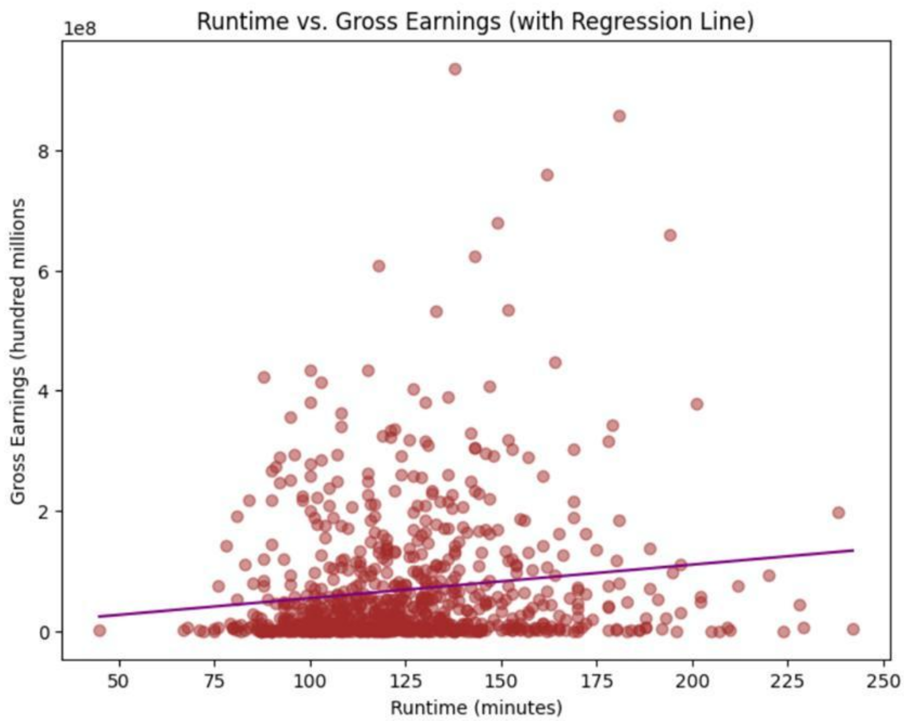

Figure 6-This is a scatter plot showing the relationship between the runtime of movies (in minutes) and their gross earnings (in hundred millions), with a regression line fitted.

Conclusion

This analysis of the IMDb Top 1000 movies dataset highlights several important relationships. IMDb rating moderately correlates with audience engagement (votes) but weakly with gross earnings. The k-NN classification model demonstrates fair performance, especially for Low and High categories, though class imbalance complicates prediction of Medium-rated movies. Lastly, while runtime shows some correlation with earnings, it is not a primary driver. Overall, the data supports the view that film success results from a complex interplay of content quality, viewer engagement, and external factors.

Works Cited

"IMDB Top 1000." *Kaggle*, www.kaggle.com/datasets/omarhanyy/imdb-top-1000.